



A review on mechanical properties and deterioration mechanisms of FRP bars under severe environmental and loading conditions

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ABSTRACT

Rebar corrosion is the most common durability issue in reinforced concrete structures, and replacing steel rebar with fiber-reinforced polymer (FRP) bar can effectively resolve the corrosion issue. Furthermore, it can serve as an appropriate substitute for steel rebar due to its lightweight, high strength, and good corrosion resistance. However, the mechanical performance of FRP bars can still deteriorate under severe environmental and loading conditions. As a result, it is essential to understand the deterioration mechanisms of FRP bars and enhance their durability performance. This study first critically reviews the influences of reinforcement, resin, and bar diameter on the mechanical properties of FRP bars in alkaline solution. The degradation of the mechanical properties of FRP bars is further evaluated in water, seawater, and alkaline solution. The influences of wet-dry and freeze-thaw cycles on the FRP bar degradation are also discussed. The degradation mechanisms are analyzed considering the coupling effect of exposure conditions, temperatures, concrete wrappings, and loadings. Then, currently available prediction models for the long-term durability of FRP bars are summarized. Finally, some suggestions for future works are proposed, and concluding remarks are made. This study can provide insight into FRP bars in severe service conditions for their research and application.

1. Introduction

The chloride salts in the external environment can lead to the corrosion of steel bars in concrete, which can further be accelerated by dry-wet and freeze-thaw cycles [1]. The corrosion can result in concrete structure cracks, spalling of concrete cover, and reduction in concrete structure load-bearing capacity [2]. Corrosion of steel reinforcement is one of the most severe causes of building structural failure. Replacing steel bars with anti-corrosive reinforcement can resolve the rust-related durability issue.

Fiber-reinforced polymer (FRP) bar can be an appropriate substitute for steel bar due to its low density, high strength, fatigue resistance, and entirely corrosion-free nature. FRP bar is made from continuous fiber as

reinforcement and polymer resin as a matrix using forming, infiltrating, and curing processes. The typical volume contents for continuous fiber, resin, and filler additives and protective materials are in the range of 40%–80%, 20%–60%, and <5%, respectively. The commonly used continuous fibers include glass, carbon, aramid, basalt, and ultra-high molecular weight polyethylene. The widely used polymer resins are epoxy (epoxy ester), vinyl (vinyl ester), and polyurethane (polyester ester). Glass fiber-reinforced polymer (GFRP), carbon fiber-reinforced polymer (CFRP), aramid fiber-reinforced polymer (AFRP), and basalt fiber-reinforced polymer (BFRP) are the most common types of FRP bars. In addition, hybrid fiber-reinforced polymer (HFRP) and steel-FRP composite bars (SFCB) are used as reinforcement [3], as shown in Fig. 1. GFRP bars are widely used in civil engineering due to the advantages of

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a vast source of raw materials, low production costs, and mature production technology. CFRP has great potential because of the excellent mechanical properties and corrosion resistance of carbon fiber. However, the high cost of carbon fiber limits its wide application, and CFRP is generally utilized for critical structural component reinforcement [4], structure rehabilitation [5], seismic retrofitting [6], and load-bearing in cable-stayed bridges [7]. The recently emerged BFRP bar has comparable mechanical properties to GFRP, and the environmentally friendly basalt fibers are produced from melted basalt rocks [8–10]. Thus, BFRP offers a wide range of application prospects and has attracted huge attention in civil engineering. The utilization of AFRP in civil engineering is rarely used because of its high hydrophilic nature and poor durability. HFRP bars are often made by combining glass fiber or basalt fiber and carbon fiber to address the aging problem of traditional fibers in humid and hot environments, resulting in excellent strength and corrosion resistance. However, the properties of HFRP bars are not completely known and are still under investigation [11–13].

The failure of a concrete structure occurs due to coupled environmental factors, and structure failure mechanisms vary depending on the environmental conditions. The coastal structure can easily be deteriorated by the chloride salts, wave erosion, and freeze-thaw cycles in the marine environment [17,18]. These environmental effects may easily cause the failure of building structures. In addition, there are significant freeze-thaw cycles in cold regions; the water inside the concrete generates reciprocal expansion force, inducing concrete cracks and major structural damage [19]. Building exposed to fire can significantly exacerbate the concrete damage due to high temperature, which may even lead to catastrophic failure of the building structure [20]. Moreover, the concrete structure may suffer from the coupled action of loads and harsh environments throughout its service life, which stimulates the creation of micro-cracks. The cracks then allow corrosive substances to enter the interior, accelerating environmental action, resulting in reinforcing damage, and ultimately causing the failure of building structures [21]. Fig. 2 shows different environmental actions during the service life of a building structure.

The corrosive-free FRP bars can still be deteriorated by exposure to a severe environment. The tensile strength of GFRP bars decreases due to water-induced damage to the fiber-matrix interface [22–24]. The deterioration rate of reinforcement accelerates with the exposure temperature. In addition to freshwater, the corrosion of FRP structures in seawater has also attracted widespread attention. For example, Wood et al. [25] soaked GFRP and CFRP materials in seawater to study their

interface strength and found that damage occurs at the boundary of the resin-rich zone. Seawater exposure can significantly lower the strength of CFRP material, but the influence on the modulus is insignificant [26]. The study by Li et al. [27] indicated that the water absorption and deterioration rate of CFRP composites immersed in artificial seawater increase with the exposure temperature. The alkaline pore solution can further accelerate the deterioration of the FRP bar in concrete, particularly at the FRP bar fiber-resin interface [28]. FRP bars are temperature sensitive and easily affected by fire, and their mechanical properties reduce substantially at high temperatures [29]. Furthermore, sustained loading can accelerate FRP bar degradation when exposed to harsh environments [30,31]. Additionally, the existing research on the deterioration of FRP bars is mainly focused on short-term laboratory results, necessitating the development of more appropriate models for the long-term prediction of FRP bars.

Therefore, studying the degradation mechanisms of FRP bars under different environmental effects is important for durable construction. This paper analyzes the deterioration of mechanical properties and degradation mechanisms of FRP bars under various environmental conditions. The current knowledge gaps on the durability of FRP bars are then identified. Finally, potential further research directions are pointed out, and key findings are summarized.

2. Mechanical properties of FRP bars and their deterioration under environmental actions

The investigation of the mechanical properties of FRP bars includes characterization of tensile strength, flexural strength, interlaminar shear strength, transverse shear strength, and bond strength with concrete matrix [14]. The degradation of FRP bars in various environments is the basis for analyzing the durability of FRP bar-reinforced concrete structures [32]. FRP bar mechanical characteristics can be influenced by various parameters, including fiber type, resin type, bar diameter, and service condition. This section provides an overview of current studies on reinforcing performance and the impact of the external environment on the FRP bar.

2.1. Bare FRP bars immersed in solution environment

2.1.1. Alkali solution

(1) Alkalinity

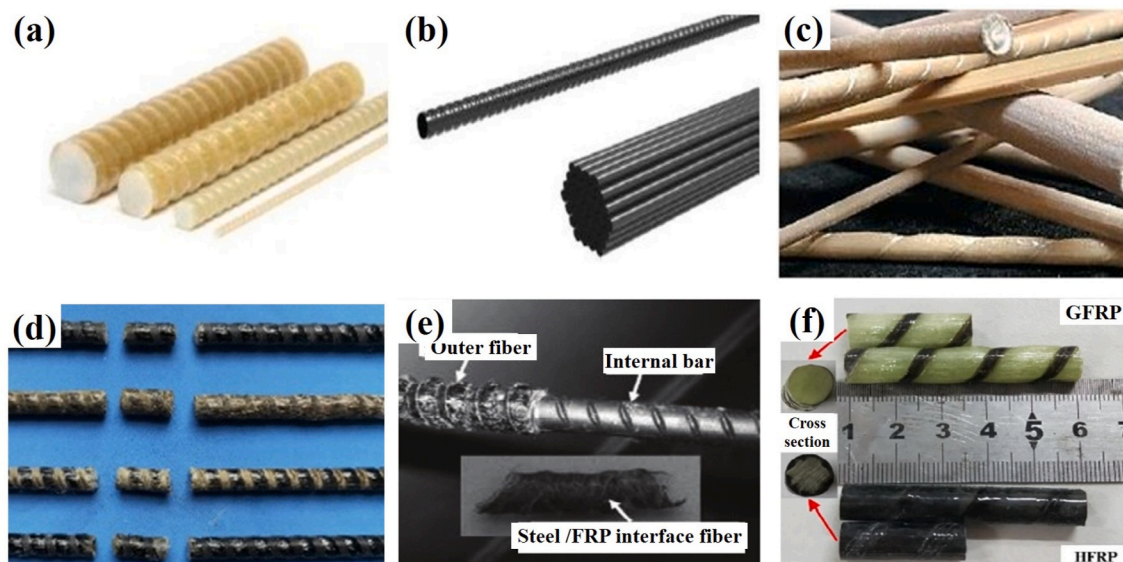


Fig. 1. Various FRP bars: (a) GFRP [14], (b) CFRP [14], (c) AFRP [14], (d) BFRP [15], (e) SFRP [16], and (f) HFRP [11].

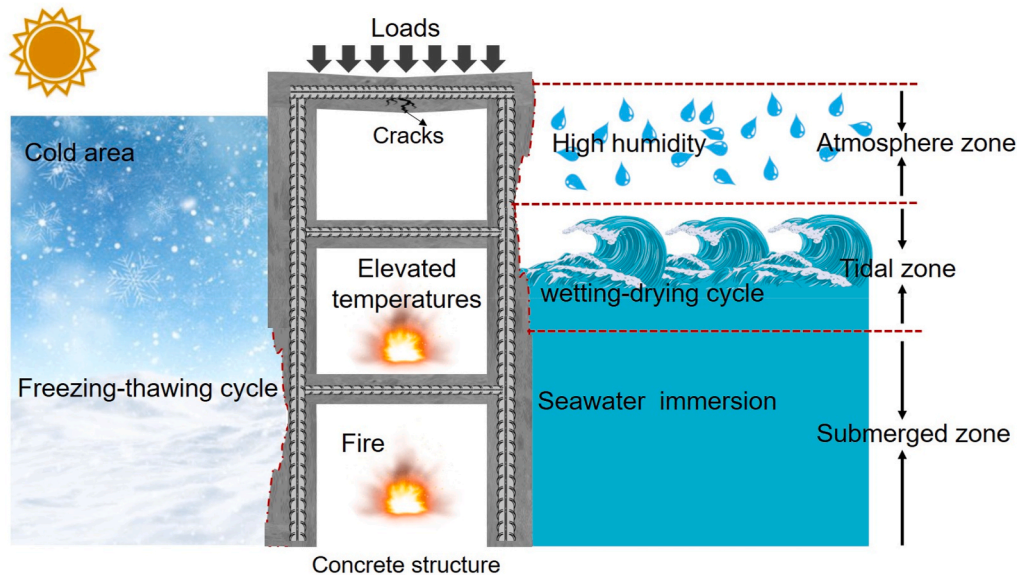


Fig. 2. Main environmental effects on a concrete structure during service.

Alkalinity is a key factor influencing the deterioration of FRP bars. The effect of alkalinity on the degradation of GFRP bars was investigated by D'Antino et al. [33,34]. The time-dependent reductions in the tensile strength retention rate of GFRP bars under different exposure conditions are summarized in Fig. 3. The impact of pH on corrosion rate is also shown in Fig. 3, with three temperature ranges of 20–32 °C, 40 °C, and 55–80 °C. The degradation of mechanical properties of GFRP bars is significantly higher at pHs of 13.4–13.6 than pHs of 12.6–12.7 (see the results reported by Chen et al. [35,36] and Wang et al. [37] in Fig. 3). In addition, higher deterioration for GFRP bars can be observed under exposure to 80 °C compared to 20 °C (Fig. 3(c)). However, with the same conditions (240 days, pH of 12.7, and 60 °C), a lower residual strength rate (59%) of GFRP bars is found in Ref. [36] than (73%) in Ref. [38]. The fluctuation in the retention rate can be caused by the non-complete bar failure [33].

The alkali ion penetration rate into the matrix of the FRP bar depends on the alkali concentration, alkali solution composition, exposure time,

and immersion temperature. The diffusion and erosion rates of alkali ions increase as alkalinity rises [15]. Notably, increasing temperature significantly enhances the diffusion rate of corrosive ions into the matrix.

(2) Fiber types

The tensile strength of FRP bars is mainly controlled by the properties of fibers. FRP bar characteristics vary depending on the type of fiber used. Benmokrane et al. [39] studied the durability of several bars (GFRP, BFRP, and CFRP) in an alkali environment and showed that carbon fiber-based bar has the maximum alkali corrosion resistance and tensile strength retention rate. SEM images of different fibers after soaking in an alkaline solution at 60 °C for 168 h are shown in Fig. 4. Carbon fibers exhibit appropriate resistance to alkali corrosion in the simulated concrete pore solution, even when CFRP bars contain microcracks in the matrix [40]. The tensile strengths of CFRP and GFRP

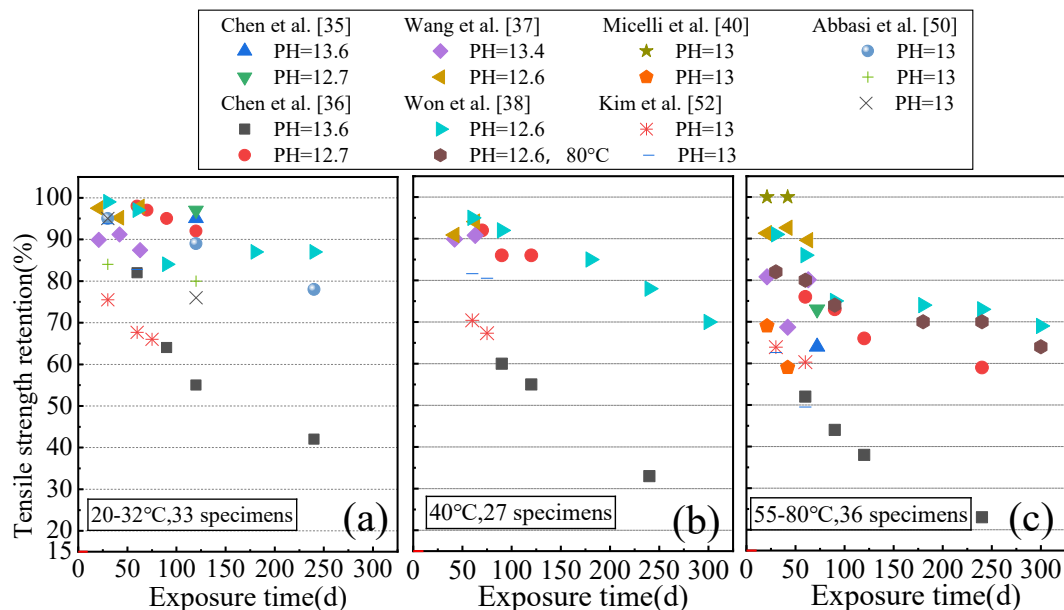


Fig. 3. Variation in tensile strength rate of GFRP bars with corrosion age under varying alkalinities: (a) 20–32 °C, (b) 40 °C, and (c) 55–80 °C.

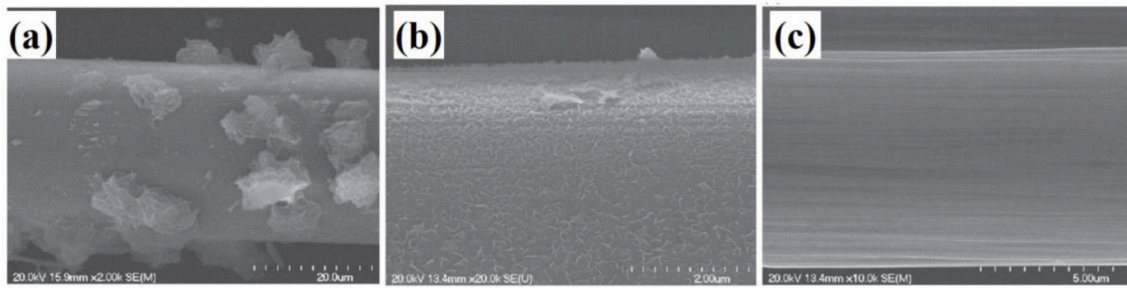


Fig. 4. Micrographs of different fibers immersed in an alkaline solution at 60 °C for 168 h: (a) glass fiber, (b) basalt fiber, and (c) carbon fiber [39].

bars decrease by 4% and 51%, respectively, after alkali solution immersion at 60 °C for 70 days [35]. When immersed in an alkaline solution at 20 °C with a pH of 12.7 for 30 days and 120 days, the tensile strengths of GFRP bars drop by 2% and 10%, respectively, as stated in Ref. [36]. GFRP bars also outperform BFRP bars in alkali solutions with pH 12.7 and 13.4 at 55 °C [37].

CFRP bars offer excellent mechanical properties and durability [40–42]. However, the high cost of CFRP bars restricts their use in the concrete industry. GFRP bars are successfully applied to roads, bridges, and other applications, although they are highly susceptible to deterioration in alkaline environments [43], similar to BFRP bars [44].

Some studies focus on hybrid fibers composed of carbon fiber and glass fiber or basalt fiber [13,45,46], as shown in Fig. 1(e) and (f). You et al. [47] compared the tensile properties of hybrid and non-hybrid fiber bars and revealed that the ultimate strain of hybrid bars increases by more than 33%. The carbon fiber can protect the inner glass fiber from erosion. The protection can prohibit the entrance of water and corrosive ions and thus enhance durability performance [11]. BFRP protective layer can also effectively protect the inner GFRP bars from acid and alkali corrosion, resulting in the preservation of the tensile properties of the bar [48]. The actual corrosion rate (estimated by micro CT scanning technology (Fig. 5(a–b)) of SFCBs is significantly less than that of ordinary steel bars; the actual corrosion rates of CFRP-SFCBs and GFRP-SFCBs are less than 1/10 and 1/100, respectively, of the corrosion rate of steel bar (Fig. 5 (c)) [49]. BFRP bars have significantly lower mechanical properties than HFRP (i.e., B/CFRP) bars [12].

To sum up, CFRP bars have better durability performance than GFRP and BFRP bars. Utilizing hybrid fibers to create FRP bars can offer superior mechanical properties and durability than using one type of fiber. Notably, the mechanical properties of hybrid fibers are also affected by fiber type, fiber surface treatment, and manufacturing technique.

(3) Resin types

The resin properties influence the degradation rate of the reinforcement [50] and greatly affect the reinforcement durability in the

building structure [40]. Polyester, vinyl ester, and epoxy are commonly used resins. Polyester resins are susceptible to alkali ion erosion, while thermoplastic resins have better durability and toughness but lower viscosity [40]. Polyester resin is also the least resistant to temperature and chemical erosion, but vinyl ester and epoxy are more weather-resistant [50]. Another study [51] found that when epoxy and vinyl ester resins are used as substrates, GFRP bars immersed in an alkali solution at 60 °C for 5000 h show a higher tensile strength retention rate than polyester resin substrates. Epoxy and vinyl ester resins have greater interfacial adhesion with glass fibers than polyester resins (see Fig. 6), which results in higher mechanical properties in epoxy- and vinyl ester-based GFRP bars than polyester-based GFRP bars. The tensile strength retention rate of GFRP bars with modified vinyl ester resin is higher than that of GFRP bars soaked in an alkaline solution at 20, 40, and 80 °C [52]. After conditioning in an alkaline solution, glass/vinyl ester FRP bars have the maximum physical and mechanical properties and minimum degradation rate, followed by basalt/epoxy FRP bars, while the basalt/vinyl ester FRP bars show the minimum physical and mechanical properties and exhibit substantial degradation of their properties [39,53].

Various fibers, surface treatments, and matrices significantly affect the mechanical properties and durability of reinforcing materials. Resin can be modified to improve weather resistance, which is an important aspect of enhancing the durability of FRP bars. For instance, modified acrylic resin is widely used in the interior decorating of automobiles. The modified acrylic resin-based GFRP bars retain the designed mechanical properties, and the tensile properties of the modified acrylic resin remain over 90% following immersion in an alkaline solution [54]. Moreover, it can provide a greater safety margin and increase escape time in the event of a fire. Newly developed thermoplastic-based GFRP bars are shown to be very durable, with tensile strength remaining about 87% after 90 days of soaking at 60 °C in an alkaline solution [55].

(4) FRP bar diameter

The size effect on the long-term durability of the FRP bar has been

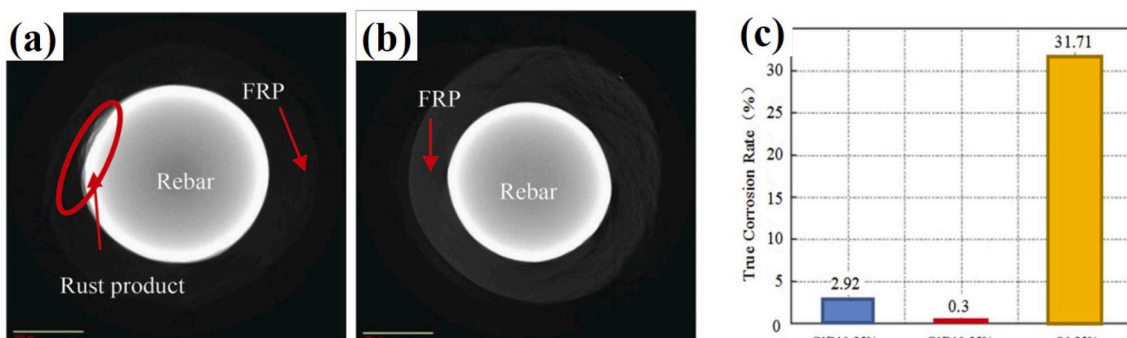


Fig. 5. Micro CT scanning of the two-dimensional cross-section of the samples: (a) CFRP-SFCBS and (b) GFRP-SFCBS, and (c) corrosion rate of different reinforcement materials [49].

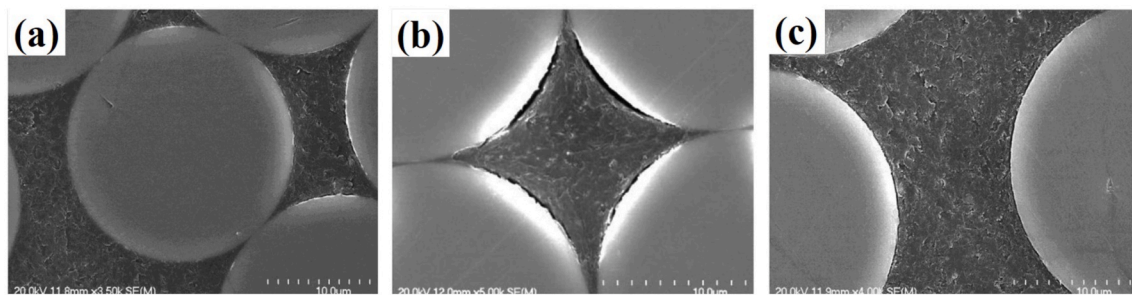


Fig. 6. Fiber-resin interfaces of GFRP bar after soaking in an alkaline solution [51]: (a) epoxy resin, (b) polyester resin, and (c) vinyl ester resin.

reported in Refs. [32,56,57]. In general, the FRP bars with a larger diameter show higher strength retention than smaller diameters. However, after soaking in an alkaline solution, the tensile strength retention rate of the 25 mm BFRP bar is lower than the 5 mm bar because the outer fiber is exposed to increased stress-induced failure, resulting in an instant propagation of the breakage to the inner (known as the shear lag effect) [32]. Similar results were obtained with GFRP bars [58]. In contrast, larger diameter bars exhibit a slower degradation of flexural strength, transverse shear strength, and interlaminar shear strength than small diameter bars in high-temperature alkaline solution [32,58]. This is due to the lower outer fiber corrosion depth of larger diameter FRP bars in the alkaline solution (Table 1) and strong fiber-resin interfacial bonding [58,59].

2.1.2. A comparison of water, seawater, and alkali solution environment

Fig. 7 displays the degradation of the tensile properties of GFRP bars with soaking time in tap water (TW), seawater (SW), and alkaline (ALK) environments at various temperatures (20–25 °C, 40–50 °C, and 60–80 °C). The mechanical properties of FRP bars degrade more significantly when the temperature rises from 20 °C to 80 °C. Also, the tensile property of GFRP bars degrades faster in the alkaline solution than in the salt solution or tap water. Similar findings were reported in Refs. [48,60,61]. Exposure to an alkaline environment can typically lead to a higher degradation of the mechanical properties of FRP bars [33, 34]. In this environment, OH^- reacts with resin, destroying the macromolecular chain and leading to its dissolution. Water molecules and corrosive ions then quickly penetrate the fiber-matrix interface through microcracks, thus causing rapid aging and reducing tensile strength [38].

In general, abundant Cl^- in seawater can cause significant corrosion of steel bars. In contrast to steel bars, Salloum et al. [62] showed that after 540 days of immersion in seawater at 50 °C, the tensile strength retention rate of GFRP bars is 11% higher than that in tap water (Fig. 7 (b)). Additionally, the tensile strength retention rate of GFRP bars soaked in seawater at 80 °C for 132 days is 4.7% higher than those in tap water (Fig. 7(c)) [52]. Although the deterioration mechanism of FRP bars in seawater is similar to that in tap water, the high salt content of seawater inhibits water diffusion into the matrix [63], as determined by the mass change before and after water absorption [64]. In addition, a very thin salt layer may form on the surface of FRP bars in seawater, particularly at higher temperatures. This salt layer greatly hinders water molecules from penetrating the FRP bar interior. Hence, the durability

of FRP bars in seawater is superior to those in tap water at higher temperatures [62].

2.2. Deterioration of FRP bars embedded in concrete

The accelerated degradation of FRP bars in concrete can represent the real deterioration in field application compared to the simulated concrete pore solution exposure. The GFRP bars wrapped in moist concrete show better durability than bare bars in the simulated concrete pore solution [65]. Manalo et al. [64] observed similar results. These results confirm that the mechanical properties of GFRP degrade at a slower rate in the concrete wrapping environment compared to those directly immersed in the alkaline solution. Several studies also compared the durability of concrete-wrapped FRP bars in seawater to tap water. GFRP bars wrapped in concrete exhibit a higher tensile strength retention rate in seawater than tap water [66,67] which is similar to the results of bare bars in section 2.2. Another study [68] obtained the opposite result. They indicated that sodium chloride in seawater penetrates concrete pores and alters the internal chemical composition, resulting in OH^- enrichment and a rise in alkalinity, which negatively impacts tensile behavior.

Only a few experimental investigations have been conducted on the durability of concrete-wrapped BFRP bars. Lu et al. [69] studied the durability of concrete BFRP and bare bars immersed in seawater at 60 °C for one year. The concrete-coated BFRP bar is more susceptible to corrosion in seawater than the bare bar, owing to the alkalinity in the concrete. Moreover, alkali in the concrete reacts with SiO_2 in BFRP bars (i.e., alkali-aggregate reaction). A white gel forms at the fiber-resin interface, contributing to a reduction in BFRP mechanical properties [70]. The tensile characteristics of BFRP bars wrapped in coral aggregate concrete degrade faster than those of ordinary concrete [71]. This is because coral aggregate contains numerous pores, leading to low permeability resistance and easy water access from the surrounding environment.

2.2.1. Concrete cover thickness

The thicker the protective layer, the more stable the passivation layer and the greater the corrosion resistance of reinforced concrete structures. However, the impact of the concrete cover thickness on the durability of FRP bars is yet unclear. Lu et al. [70] examined the durability of BFRP bars in seawater for 360 days by varying the thicknesses of the SWSSC mortar protective layer. It was found that the alkalinity of the mortar gradually decreases after seawater immersion because of the alkaline environment. The higher the alkalinity, the greater the damage to FRP bars. Tensile strength is lower in BFRP bars with a 20 mm SWSSC cover thickness than in BFRP bars with a 10 mm SWSSC cover thickness. They also found that SWSSC reaches alkali equilibrium after 180 days at 40 °C and 90 days at 60 °C, respectively, when the protective layer thickness is 10 mm. Moreover, ambient humidity influences the effect of concrete cover thickness on the tensile strength of GFRP bars [68]. When the relative humidity is greater than 85%, the tensile strength retention rate of FRP bars decreases with increasing protective layer thickness.

Table 1
Corroded portion of FRP bars [58].

Affected portion	Specimen No.				
	1	2	3	4	5
Nominal diameter of the bar (mm)	9.5	12.7	15.9	19.1	25.4
Area (10^{-3} mm^2)	138.2	186.2	270.0	111.7	223.0
Thickness (10^{-3} mm)	4.4	4.7	5.4	1.9	2.8
Percentage of nominal diameter (%)	0.05	0.04	0.03	0.01	0.01

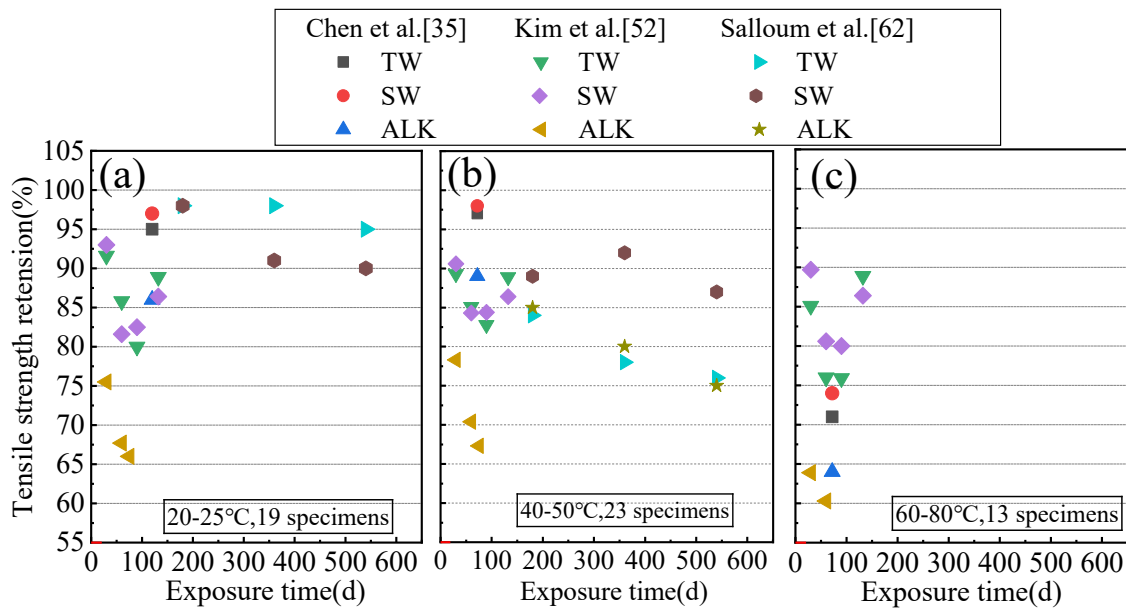


Fig. 7. Comparison of GFRP bars with corrosion age in seawater, tap water, and alkaline solution at various temperatures: (a) 20–25 °C, (b) 40–50 °C, and (c) 60–80 °C.

When the relative humidity is 75%, the concrete may prevent the GFRP bars from eroding. The tensile strength retention of 62 mm GFRP bars is higher than that of 47 mm protection bars.

2.2.2. Water-cement ratios

In general, the porosity of concrete increases as the water-cement ratio rises, and lower porosity provides better protection for steel bars [72]. Different from steel bars, after 120 days of soaking in tap water at 60 °C, the tensile strength retention of GFRP bars coated with water-cement ratios of 0.31, 0.4, and 0.6 increases as the ratio rises [68]. It can be inferred that the lower the water-cement ratio of the bar, the faster the deterioration rate and the lower the tensile strength retention rate. This is for two reasons: (1) The low water-cement ratio contains more cementitious ingredients than concrete, and more alkali ions accumulate on the reinforcing surface. (2) The low porosity of concrete with a low water-cement ratio makes it more difficult for alkali ions to migrate out, resulting in significant alkalinity in concrete with a low water-cement ratio [68].

Overall, the deterioration of FRP bars in concrete depends on the concrete cover thickness, water-cement ratio, and immersion environment. Further investigation is needed to explore their influencing mechanisms.

2.3. FRP bars subjected to temperature effect

2.3.1. Typical temperatures

The elevated temperature can accelerate the degradation of FRP bars. The tensile strength retention rate of BFRP bars is 36% lower at 60 °C than at 20 °C after soaking in an alkali solution for 1000 h [60]. The degradation of BFRP bars accelerates with increasing temperature [73]. Rifai et al. [74] investigated the tensile strength retention rate of BFRP bars soaked in an alkaline solution and covered with moist concrete at 20, 40, and 60 °C. They showed that the higher the temperature, the longer the soaking time, the stronger the alkaline effect, and the lower the tensile strength of the bars. After soaking in an alkaline solution at 60 °C for nine months, the tensile strength retention rate of BFRP bars is 25% lower than those of 20 °C, whereas at 60 °C moist concrete compared to 20 °C, the reduction is 6.5%. After 28 months of immersion in tap water and alkaline solution at 50 °C, the reductions in tensile strength of GFRP bars are 24.48% and 24.05%, respectively,

while there is almost no loss of tensile strength at room temperature [75]. However, despite degrading the tensile strength, the elastic modulus of BFRP bars increases by 6.5% on average [60]. This can be attributed to the post-curing of the resin [60,76]. It can be inferred that the higher the temperature, the faster the FRP bar deterioration and the lower the tensile strength retention rate of the FRP bar.

2.3.2. Extreme temperatures

Since high-temperature fires or freezing weather often damage FRP structures, researchers have studied their mechanical properties in high-temperature and extremely cold environments. For instance, Wang et al. [77] examined the mechanical properties of GFRP bars at temperatures ranging from 20 to 600 °C, and 350 °C is identified as a critical temperature (at which an FRP bar loses about half of its tensile strength [78]) for the mechanical properties of bars [77,79]. Below this temperature, the elastic modulus of the bars remains 90%. The mechanical properties of the FRP bar deteriorate slowly with increasing temperature, while the degradation is significant in the GFRP bar. In another study, Ashrafi et al. [78] found that the critical temperatures of GFRP bars with diameters of 4, 6, 8, and 10 mm are 300, 375, 377, and 450 °C, respectively. Spagnuolo et al. [80] showed that the tensile strength and elastic modulus of GFRP bars tend to remain stable in the temperature range of 100–300 °C, indicating that these temperatures do not affect the mechanical properties of GFRP bars. After exposure to 300–500 °C, the glass fibers fracture occurs due to a zero-stress corrosion mechanism, and the tensile strength drops significantly [80]. Surface morphologies of GFRP bars at various temperatures are illustrated in Fig. 8.

Ashrafi et al. [78] showed that in addition to temperature, FRP bar diameter, fiber type, resin type, and thermal parameters (T_g and T_d) of FRP bars have an effect on their properties, but the fiber-to-matrix ratio has a minimal impact [78]. It was shown that exposing FRP bars to high temperatures up to 450 °C reduces their ultimate tensile strength substantially, with FRP bars experiencing approximately 50–70% decline in tensile strength. The strength retention rates of 8 mm GFRP bars are 98, 81, 69, and 40% at 80, 150, 300, and 500 °C, respectively [29]. The stiffness of the amorphous polymer matrix increases in the low-temperature range of –100 °C to 0 °C, leading to a gradual increase in the tensile strength of GFRP bars as the temperature decreases [81]. When the temperature is between –40 °C and 50 °C, the tensile strength and elastic modulus of GFRP bars are unaffected by temperature. When

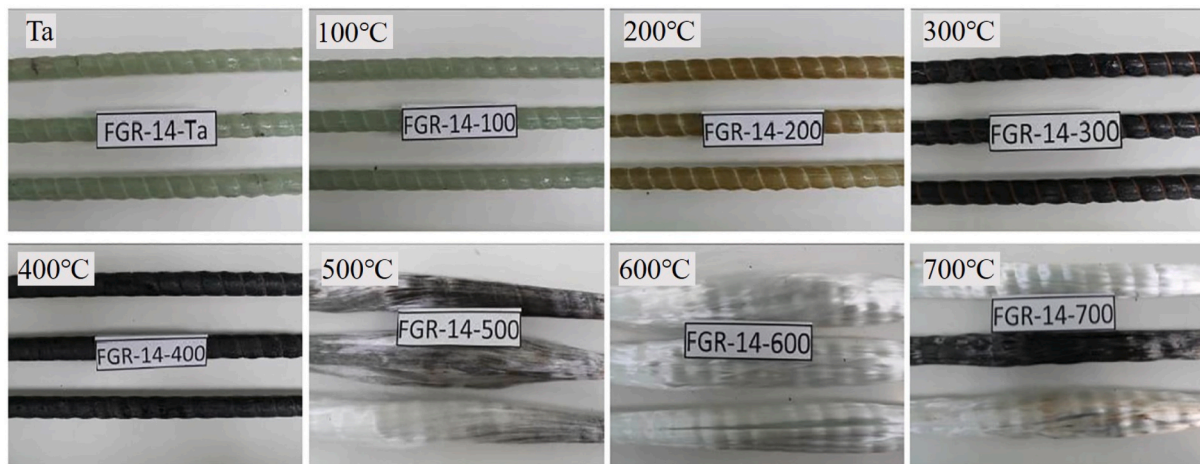


Fig. 8. Surface morphologies of GFRP bars at different temperatures [80].

the temperature exceeds T_g , the modulus and strength of FRP bars rapidly decrease due to the degradation of the polymer matrix. When the temperature is high (350 °C), the strength of FRP bars significantly decreases owing to the formation of microcracks in the matrix [81]. In addition, variations in bar diameter influence the tensile properties of FRP bars when subjected to various temperatures. In the temperature range of 150 °C–500 °C, bar diameter has a significant effect, while below 150 °C and above 500 °C, bar diameter has a little effect [29,82]. The mechanical properties of fireproof coated GFRP bars remain unchanged between room and glass transition temperatures (i.e., 25–80 °C) [83]. When the temperature is 80–120 °C, the mechanical properties of the bars decrease rapidly due to the fiber and resin debonding. At 120–350 °C, a slight decrease in the mechanical properties of the bar is observed, but the fire-retardant coating has no effect. At 350–600 °C, the fire-retardant coating effectively prevents the loss of FRP bar mechanical properties and increases the tensile strength retention rate by 20%–30%. When the temperature exceeds 600 °C, the performance of fireproof coating significantly decreases due to the extremely high temperature. The failure modes at different temperatures are depicted in Fig. 9.

It can be deduced that the mechanical properties of FRP bars deteriorate significantly at high temperatures and gradually decrease with increasing temperature. As shown in Fig. 10, the tensile strength rises to some degree between –100 °C to 0 °C. When the temperature exceeds 300 °C, the tensile properties quickly decrease. However, the high-temperature resistance of FRP bars can dramatically be improved by coating their surfaces with a protective layer or wrapping concrete.

2.4. Other environmental effects

2.4.1. Wetting and drying cycles

Structures in coastal and other environments may often be subjected to dry-wet cycles. These cycles aggravate reinforcement corrosion, cause concrete cracking and structural failure [84,85]. Aiello and Ombres [86] carried out dry-wet cycle tests on AFRP and GFRP bars immersed in water at 20 °C for two days and dried in air for two days to determine the mechanical property deterioration. After 15 cycles (60 days), the strength losses of AFRP and GFRP bars are 13.74% and 29%, respectively, and the elastic modulus losses are 7.21% and 13.29%, respectively. The degradation effect of 72-day dry-wet cycles in alkaline solution on GFRP bars is less significant than 70-days continuous immersion at 60 °C [35]. This is because, for continuous immersion, the time in the solution is relatively longer, offering water molecules and corrosive ions more time to penetrate the matrix, resulting in a decrease in the mechanical properties of the bar. The tensile strength of GFRP bars is reduced by 9.82% during the 18-month dry-wet cycles in

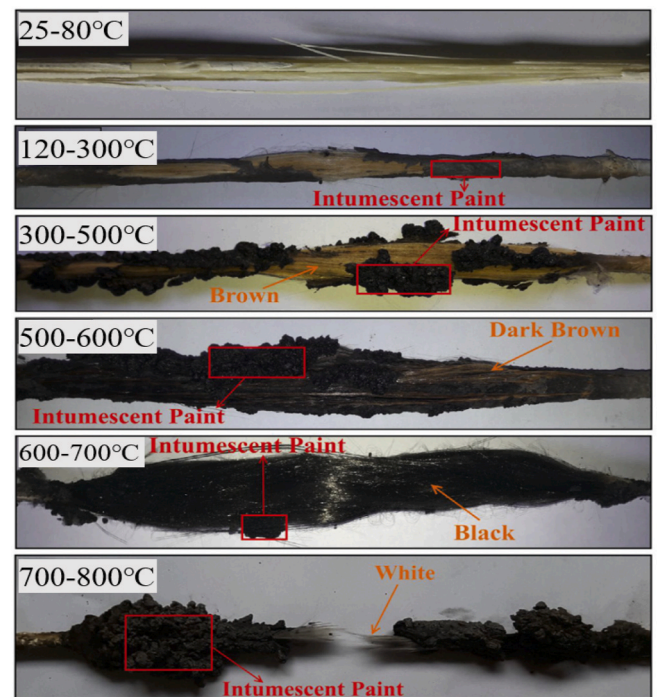


Fig. 9. Failure modes of GFRP bars coated with intumescent paint at low and high temperatures [83].

seawater at 50 °C, whereas the tensile strength loss of GFRP bars is 12.80% [75]. A similar observation was reported by Almusallam et al. [62]. In contrast, Kim et al. [52] showed that the dry-wet cycles have no discernible impact on the corrosion of FRP bars. It can be seen that the wetting and drying cycles have no obvious effect on the corrosion of FRP bars. The corrosion rate of the FRP bar is more severe under full immersion than that under drying conditions.

2.4.2. Freezing and thawing cycles

Freeze-thaw cycles in cold locations can deteriorate concrete materials and structures [87]. The tensile strength degradation of GFRP bars is minimal after 30, 60, and 130 days of freeze-thaw cycles in an alkaline solution [52]. Tannous et al. [88] carried out 1200 freeze-thaw cycles on CFRP and AFRP bars in the air, holding them at –30 °C for 2 h and subsequently at 60 °C for 2 h. The freeze-thaw cycles in the air humidity environment have no noticeable effect on the mechanical characteristics

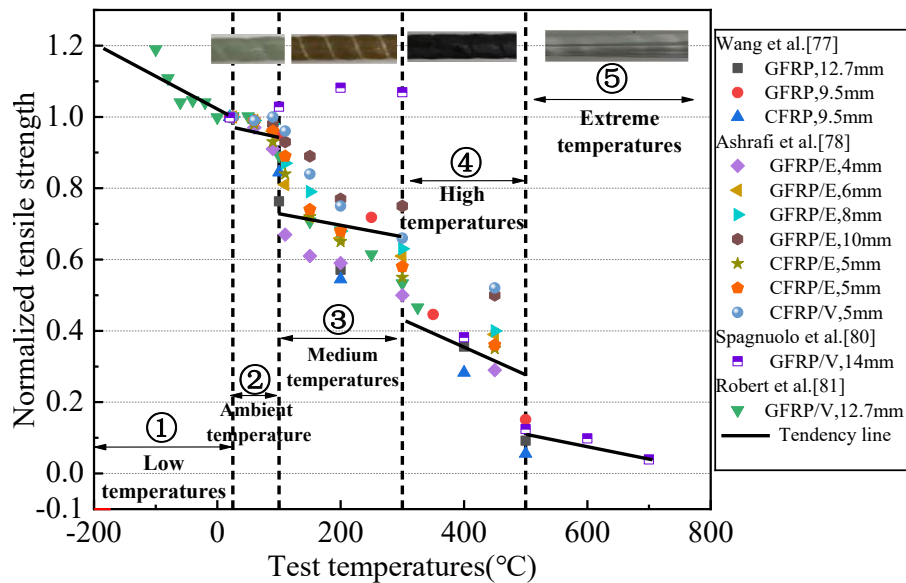


Fig. 10. Tensile properties of FRP bars at different temperatures.

of FRP bars. These cycles also have little influence on the degradation of the interlaminar shear strength of FRP bars in tap water and alkaline solution from -20°C to 20°C [89]. The tensile strength loss of the specimen subjected to freezing and thawing cycles in an alkaline solution is similar to that of an alkaline condition, indicating that alkaline erosion is the leading cause of the strength reduction in both cases [52]. It can be stated that the freeze-thaw cycle has little influence on the loss of mechanical properties of FRP bars.

2.4.3. Field exposure

The characterization of the degradation of FRP bars under field exposure can help unveil the deterioration mechanism under real service conditions. Lu et al. [69] placed BFRP bars wrapped in concrete in the outdoor seawater immersion region ($0.9\text{--}24^{\circ}\text{C}$). After one year of exposure, the reductions in tensile strength of BFRP bars wrapped in concrete and without encapsulation are 35% and 11%, respectively. The tensile strength of bare GFRP bars does not change in the high heat and high humidity outdoor environments (50°C and RH of 52%) following 18 months of exposure, while the tensile strength retention rates of concrete coated GFRP bars are 98.1% and 97.6%, respectively [62,75]. After 15 years of service, the GFRP bars of “Sierrita de la Cruz Creek Bridge” located in Texas, show no degradation, no damage to fiber and resin and fiber-resin interface, and excellent contact between concrete and GFRP bar interface [90]. No significant changes in the physico-chemical characteristics and microstructure of the concrete and vinyl ester-based GFRP bars sampled from the “Val-Alain bridge” barriers (located in Quebec) are detected after 11 years of its service exposure to wet-dry cycles, freeze-thaw cycles, and deicing salts [91]. Ramanathan et al. [92] found similar results using the GFRP bar obtained from dry-dock at Pearl Harbor in Oahu, Hawaii.

The current field exposure experimental investigations have shown that the FRP bars have excellent durability in dry conditions compared to indoor conditions, but they degrade quickly in a wet environment. However, only a few studies have looked into the degradation mechanism of reinforcement in a field exposure environment. As a result, further research into the durability of reinforcement in a field exposure environment is needed.

2.5. Combined effect of sustained load and environment

Concrete structures often suffer from the combined effects of multiple environmental conditions and sustained loads during their service

life. The sustained load has a major impact on the deterioration of FRP bars in harsh conditions. The load can cause microcracks in the matrix, and these cracks allow the environmental media to enter the matrix [93]. As a result, to ensure the performance of FRP bars throughout the service life of a typical civil infrastructure, it is necessary to study the actual performance of FRP bar concrete members under real service loads and environmental conditions. Evaluating the durability performance of FRP bar-strengthened structures under service loads and natural weathering conditions may be one of the main objectives of future FRP bar durability research [52]. The mechanical behavior (creep, relaxation, and residual tensile) degradation and deterioration mechanism of GFRP and BFRP bars when subjected to combined loads and environmental conditions are summarized in this section.

2.5.1. Methods of applying sustained loading

The test device is crucial for the reliability of the acquired experimental data [94]. There are currently no applicable guidelines and specifications for applying sustained loading, and various researchers use self-designed devices [95,96]. Two techniques are available for evaluating FRP bars under the coupled action of sustained load and environment. One approach uses the lever principle to load the FRP material (Fig. 11(a)). The testing machine, as illustrated in Fig. 11(b), is applied for the relaxation and creep tests, and the FRP bars are soaked in the corrosive solution for environmental conditioning (Fig. 11(c)). Another method is to use the steel frame or threaded rod as a reaction-supporting device. The solution immersion can be accomplished by wrapping the PVC sleeve around the FRP reinforcement (Fig. 12(a)). In addition, some researchers utilized a preloaded spring to provide a continuous load to FRP bars and then wrapped the PVC sleeve around the FRP bars for solution immersion, as shown in Fig. 12(b) [96].

2.5.2. Creep and relaxation behavior of FRP bar under sustained loading

Sustained loading may cause progressive longitudinal deformation of viscoelastic FRP bars, commonly referred to as creep [97,99]. Creep rupture occurs as a result of the accumulation of creep deformation and progressive crack opening under the sustained load [100]. Furthermore, stress relaxation can also happen when FRP materials are subjected to constant strain [101]. Creep and relaxation are important time-dependent behaviors that substantially affect FRP bars' application in civil engineering [94]. D'Antino et al. [97] revealed that long-term testing does not affect the tensile strength and elastic modulus of GFRP bars when sustained stress is less than 60% of UTS (ultimate

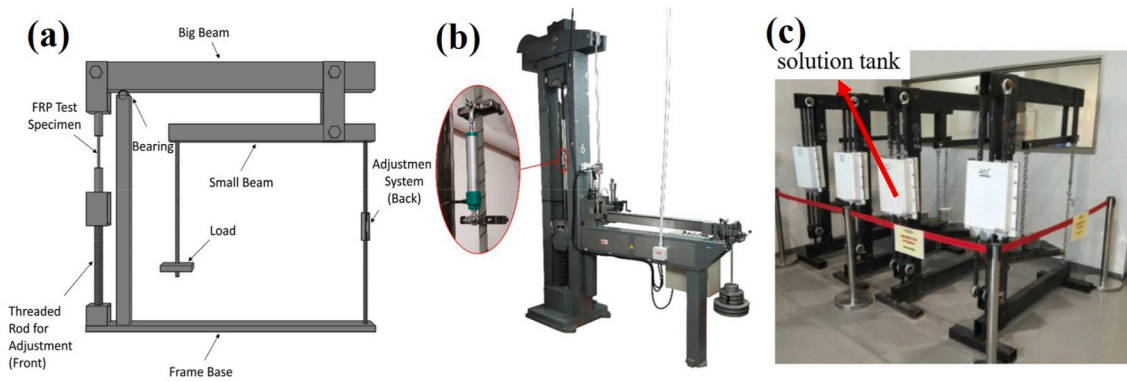


Fig. 11. Loading system: (a) elevation of the loading frame to magnify the applied load [95], (b) testing machine used for the relaxation and creep tests [97], and (c) machine used for coupled action of sustained load and solution immersion [98].

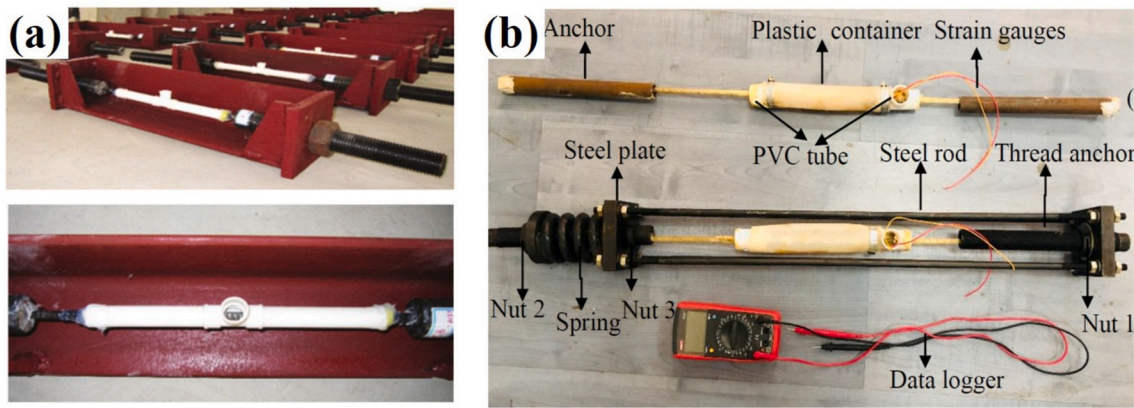


Fig. 12. Static loading devices: (a) preload nut fixation [73] and (b) spring loading [96].

tensile strength) [97]. When the sustained load of GFRP bars exceeds 30% of UTS, the ratio of ultimate creep strain to initial elastic strain decreases [102]. The creep tests on GFRP bars under sustained load (25% of UTS), seawater concrete, and varied temperatures (20, 40, and 60 °C) were performed by El-Hassan et al. [98]. They demonstrated that GFRP bars experience creep failure at higher temperatures. The predicted creep fracture strength of BFRP bars is 0.586UTS [103–105]. The fiber content also significantly affects the long-term creep properties of BFRP bars [106]. Shi et al. [105] investigated the creep behavior of BFRP bars after salt solution immersion and showed that the ratio of

creep fracture stress to the original tensile strength of BFRP bars decreases significantly. The salt solution has no effect on the relaxation behavior of BFRP bars. Under coupled action of sustained loading and alkaline solution aging at 60 °C, BFRP bars show a marginally smaller creep coefficient than other AFRP materials [95]. It can be highlighted that long-term creep and relaxation behaviors are influenced by environmental conditions (temperature, moisture, and chemical), sustained loads, and bar types (BFRP, GFRP, and AFRP).

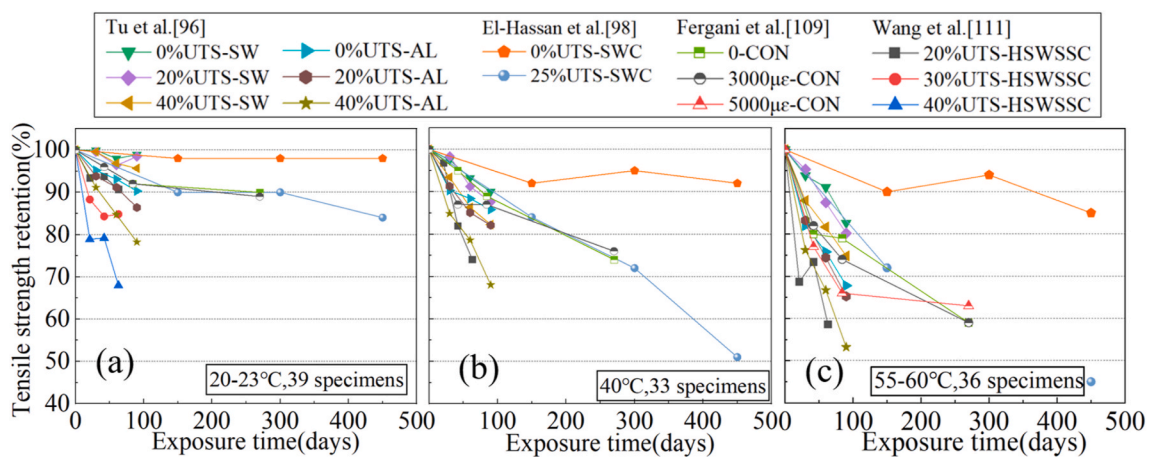


Fig. 13. Variation in the tensile strength retention rate of GFRP bars with time under the coupling action of different sustained loadings, environmental conditions, and temperatures ((a) 20–32 °C, (b) 40 °C, and (c) 55–60 °C).

2.5.3. Residual mechanical properties of FRP bars under the coupled action of sustained load and environment

(1) GFRP bars

Studies on GFRP bars were typically conducted at various environmental effects and stress levels. Fig. 13 depicts the variation in the retention rate of the tensile strength of GFRP bars as a function of exposure time under the coupled action of loads (0%–40% of UTS) and environment effects (seawater (SW), alkaline solution (ALK), seawater concrete (SWC), and concrete (CON)). The strength retention rate decreases with increasing exposure time and temperature (Fig. 13(a)–(c)). The stress level has a considerable effect on the mechanical deterioration of GFRP bars, while tensile strength retentions are about 92% and 67% (25 °C, ALK, and 63d) for loadings of 20% of UTS and 40% of UTS, respectively (Fig. 13(a)). Similar results were observed by Tu et al. (Fig. 13(b)) [96]. Mohammad et al. [76] found no significant difference in the degradation rate of GFRP bars at the ultimate stress level of 30%. The stress level of GFRP bars should be restricted to less than 25% of the standard design strength (i.e., average ultimate tensile strength minus three times the standard deviation) [28]. Nkurunziza et al. [107] studied the durability of GFRP bars under the coupled action of different sustained tensile loadings (25% and 38% of UTS) and environment factors (tap water and alkaline solution). After soaking for 10000 h (i.e., 417 days), GFRP bars displayed good durability under the coupling action of load and environment. There was no notable difference in the elastic modulus of the FRP bars under the stress levels and environmental conditions used. The compressive strength retention rate of GFRP bars soaked in salt and alkaline solutions gradually decreases with increasing sustained stress levels (0, 20, and 40% of the ultimate compressive strength), exposure temperature (40, 60, and 80 °C), and exposure duration (16, 60, and 90 days) [108]. The tensile properties of GFRP bars under the sustained stresses of 3000 $\mu\epsilon$ and 5000 $\mu\epsilon$, and various environmental conditions (moist concrete, alkaline solution, and tap water) were evaluated by Fergani et al. [109]. When GFRP bars are subjected to 3000 $\mu\epsilon$ of continuous stress, their tensile strength retention is always within limits recommended in the Canadian High Durability Reinforcement Code for the bar, whereas lower average strength retention is found for higher levels of sustained stress of 5000 $\mu\epsilon$ (see Fig. 13 (c)).

Overall, the coupling effect of sustained load and environment accelerates the degradation of FRP bars. When the load level is below 30% of the ultimate load, the continuous load has little influence on the tensile properties of the FRP bar, while when it surpasses 30% of the ultimate load, micro-cracks develop in the resin, resulting in a substantial rise in the degradation rate of FRP bars. However, the limit of sustained load stipulated by the American Concrete Association is 14% of ultimate strength, which is somewhat conservative [98,110]. Furthermore, the degradation rate of the FRP bars accelerates with increasing temperature. When the annual average temperature is less than 20 °C, the continuous load limit can be raised to 25% of the ultimate strength. For the annual average temperature exceeding 20 °C, the durability of GFRP bars is inadequate to sustain 25% of the continuous load [85]. In addition, since the FRP bars are directly immersed in the corrosive solution, the prediction results based on the Arrhenius formula and the field test are rather conservative [78]. Existing models for predicting deflection and crack width do not consider the coupling impact of the harsh environment and sustained load [84]. So far, there is no prediction model considering the coupling effect of compressive stress and ambient temperature [108].

(2) BFRP bars

Currently, there is limited research on the durability of BFRP bars under the coupling action of load and environment. The variation in the tensile strength retention rate of BFRP bars with exposure time under the

coupled action of different sustained loads and environments is illustrated in Fig. 14. Wu et al. [73] explored the tensile properties of BFRP bars under the coupling action of different loads (20, 40, and 60% of UTS) and simulated harsh environments (alkaline solution, salt solution, acid solution, and de-ionized water). When the stress is less than 20% of the ultimate strength, it has a minimal effect on the tensile strength of bars. When the stress level is 40% of the ultimate strength, the deterioration degree of the bar significantly increases. When the stress level reaches 60% of the ultimate stress, the bar failure is primarily controlled by creep fracture. Similar results were obtained by Wang et al. [111] (see Fig. 14). BFRP bars have excellent corrosion resistance to a salt solution. The tensile strength degradation rate exhibits a nonlinear relationship with the prestressing, while the elastic modulus has no obvious relationship with the prestressing and soaking time, and remains constant [112].

The synergistic effect of sustained load and environment has a considerable influence on the tensile strength of BFRP and GFRP bars but has little influence on their elastic modulus [111]. This is because the tensile modulus is mainly determined by the fibers and is less affected by resin and fiber-matrix interface deterioration [69]. It should be pointed out that the deterioration of BFRP bars is nonlinear with loadings, and the load mainly affects the tensile strength of the bars while having a minimal effect on their tensile elastic modulus. In addition, some researchers focused on BFRP bars soaked in an aqueous solution, which is quite different from the real concrete internal environment. Therefore, further studies based on the coupled action of load and real concrete internal environment should be conducted to obtain a more realistic degradation mechanism of BFRP bars.

2.5.4. Degradation mechanisms of FRP bars under coupled action of sustained load and environment

External factors such as physical load and environmental actions are the primary causes of FRP bars degradation under sustained loads and environmental conditions. FRP bars deterioration varies depending on the sustained load levels. Fig. 15 illustrates the degradation mechanisms of FRP bars subjected to coupled sustained loads and environmental actions. Ion diffusion, crack propagation, and stress all play an important role in the corrosion of FRP bars under different loads (see Fig. 15). When the stress level is low (25%–30% of ultimate strength), the resin shows viscoelasticity. Loads have a negligible effect on FRP bars; FRP bars are mainly affected by the diffusion rate of corrosive ions (H_2O and OH^-) of the solution. The corrosion of FRP bars is mostly influenced by crack propagation at ultimate strength levels ranging from 30% to 50% [96]. Continuous stresses in the matrix promote the formation and expansion of microcracks, causing the corrosive ions to erode and destroy the fiber-matrix interface and thus lower the tensile strength of

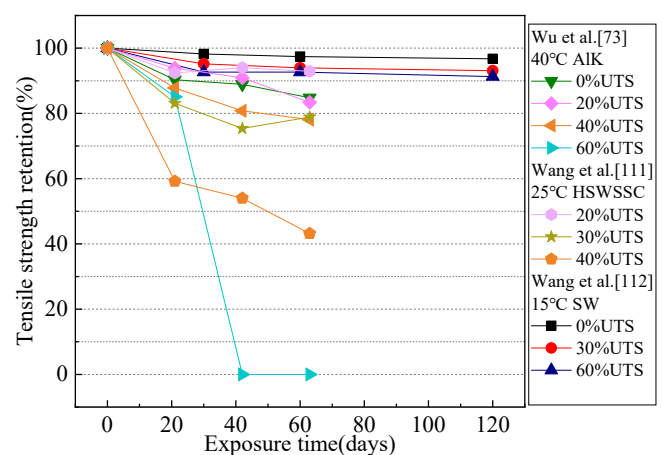


Fig. 14. Tensile strength retention of BFRP bars with different exposure times under the coupled action of varying sustained loads and environments.

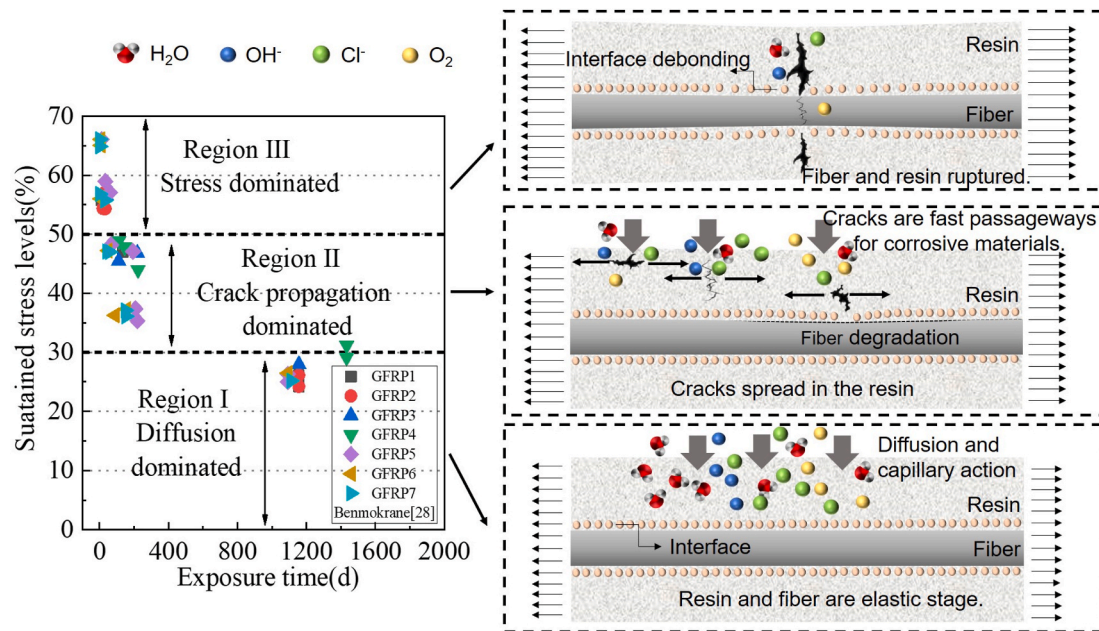


Fig. 15. Schematic diagram of degradation mechanisms of FRP bars under coupled action of sustained loads and environments.

the bar. When the load level is between 50% and 65%, the stress dominates the damage, and brittle fractures can occur quickly [73]. Additionally, stress corrosion theory containing a shear lag model, simulation, and a probability model for flaw size can explain the degradation mechanism, as shown in Ref. [113]. It can be concluded that the fracture of the FRP bars under sustained load and environmental factors is related to the stress level and the concentration of aggressive ions in the environment, as well as the manufacturing quality of the FRP bars, the property of the resin matrix, and the exposure time.

2.6. Summary

Both external environmental factors and internal parameters affect the deterioration process of FRP bars for mechanical characteristics and durability (Fig. 16). The corrosion of the alkaline solution is the most severe cause of FRP bar failure. Hydrolysis is the primary cause of the deterioration of FRP bars in seawater and tap water. However, the water absorption rate of FRP bars decreases owing to the high salt content in seawater, and the durability of FRP bars is slightly higher in seawater than in tap water.

The FRP bar wrapped in concrete is more consistent with the natural environment. FRP bars in moist concrete have higher tensile strength

than FRP bars soaked in an alkaline solution. In addition, the failure patterns of FRP bars are affected by the concrete cover thickness, and the water-cement ratio is different from that of steel bars, but the main cause of the failure of FRP bars is the alkalinity inside the concrete. When the temperature ranges from $-100\text{ }^{\circ}\text{C}$ to $0\text{ }^{\circ}\text{C}$, the mechanical strength of FRP bars slightly improves, and when the temperature is higher than $300\text{ }^{\circ}\text{C}$, the tensile strength drops rapidly.

Dry-wet and freeze-thaw cycles have a minimal effect on the durability of FRP bars, and deterioration is influenced by humidity if the exposure duration is long enough. The coupled action of sustained load and environment represents the natural environment. When the load is below 30% of ultimate strength, it has little influence on the durability and tensile strength of FRP bars. When the load is higher than 50%, the mechanical properties decrease rapidly.

FRP bars deteriorate slowly in a field environment and have good durability in a dry environment. The environmental action is the external factor that may cause the reinforcement to fail, but the ability of the reinforcement to resist the environmental action is crucial. The HFRP bars have excellent durability, and resin modification can effectively improve the weatherability of the reinforcement matrix, resulting in enhancing the durability of the bar.

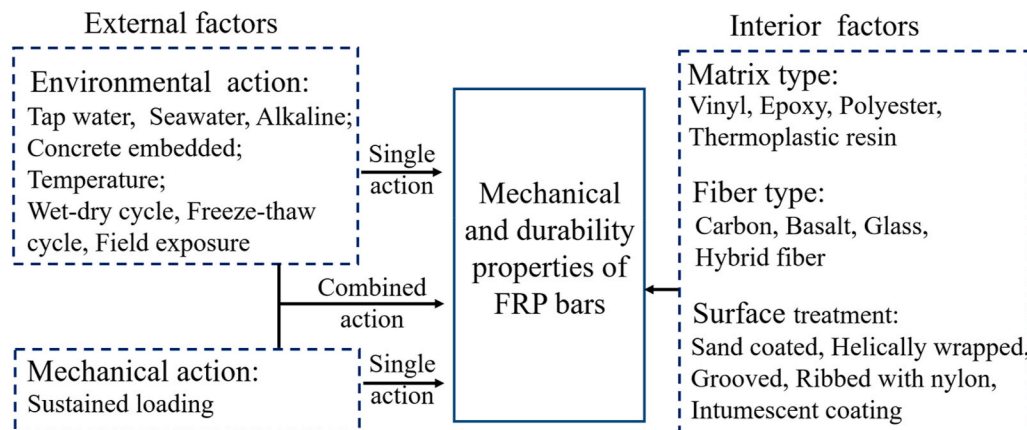


Fig. 16. Factors influencing the mechanical properties and durability of FRP bars under various environmental actions.

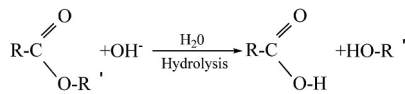
3. Deterioration mechanisms of FRP bars

FRP-reinforced concrete structures inevitably deteriorate in their service life due to highly alkaline or humid and hot environments. Therefore, analysis of the FRP bar degradation mechanism is of great importance for its long-term design and technical guidance. Degradation of fiber, resin, and fiber-resin interface is the primary cause of FRP bar durability failure. The corresponding degradation mechanisms are discussed in this section.

3.1. Deterioration mechanism of resins

Resin degradation is mainly caused by plasticization and hydrolysis reactions [114]. The water absorbed in the resin matrix acts as a plasticizer, which lowers the glass transition temperature (T_g) of the resin and reduces its flexibility [115]. As plasticization is a physical change, mechanical properties of the resin matrix recover partly after the volatilization of water molecules [116], which is known as post-curing. In addition, the water absorbed in the resin undergoes an irreversible chemical process called hydrolysis reaction, which damages the segmental chain of the resin molecules and reduces the bonding force significantly between molecules [117]. Resin hydrolysis is the main cause of a reduction in the tensile properties of FRP materials [117,118].

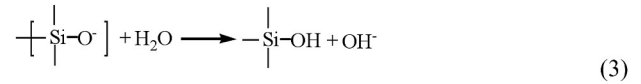
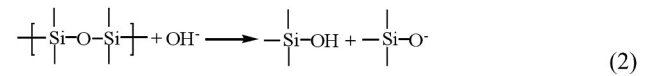
The concrete interior is highly alkaline; abundant free OH^- produces severe resin degradation. The resin matrix is etched in an alkaline environment, and the lipid bond is the weakest region in the macromolecular segment chain, such as polyester and vinyl ester. In this environment, OH^- easily destroys and hydrolyzes the ester group [119,120]. The reaction Equation is shown in (1). The content of lipid bonds in vinyl resins is far less than that of polyester resins, which is why the hydrolysis and alkalinity resistances of vinyl resins are better than that of unsaturated polyester resins. In addition, the conversion rate of vinyl ester resins is higher than that of polyester, resulting in a compact microstructure [121,122]. Fig. 17(a) shows the SEM images of resin degradation.



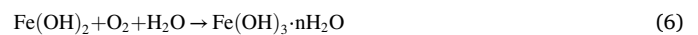
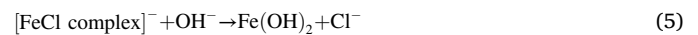
3.2. Deterioration mechanism of fibers

Since the chemical composition of carbon fibers is stable in the alkaline solution, the degradation mechanisms of glass and basalt fibers are only discussed here. Glass and basalt fibers are compounds formed by the Si–O–Si main chain structure, and the main chain segment of fiber can severely be destroyed by OH^- etching in an alkaline environment [35]. Given that basalt and glass fibers have great similarities in composition and structure, the degradation of basalt fibers in an alkaline environment may also be mainly attributed to OH^- etching [123]. The

main chain structure reacts with hydroxide ions to form solid Si–OH and liquid SiO^- , and SiO^- further reacts with water to finally form Si–OH. The reaction Equations are shown in (2) and (3), respectively.



In addition, glass and basalt fibers contain a small amount of Al and Mg oxides. As expressed in Equations (4)–(6), metal oxides react with OH^- to generate soluble substances, which then react with Cl^- in seawater to generate soluble chloride oxide [63]. It is worth noting that basalt fiber also contains about 8.51% of the total weight of Fe_2O_3 . Fe^{3+} reacts with OH^- to produce $\text{Fe}(\text{OH})_3$, which is twice the volume of Fe_2O_3 . The expansion force is generated to accelerate the deterioration of basalt fiber [63]. In addition, the divalent iron oxides contained in the fibers react with H_2O , O_2 , and Cl^- , eventually producing rust (i.e., $\text{Fe}_2\text{O}_3 \cdot n\text{H}_2\text{O}$) [124]. The agglomeration of rust in the fiber-resin interface absorbs a large number of water molecules, which causes the expansion of the resin, resulting in internal cracks in the BFRP bars under tensile conditions [37]. Fig. 17(b) shows the SEM images of fiber degradation.



3.3. Degradation mechanism of the fiber-resin interface

The fiber-resin interface is a major cause of mechanical property degradation of reinforcement. Unlike steel bar (i.e., isotropic material), FRP bar is an anisotropic material in which the fiber and matrix cooperate longitudinally, and the interface is easy to debond during the stress process. The fiber-matrix interface is about a 1 μm thick inhomogeneous region prone to deterioration due to inferior decomposition. Matrix permeability cracking, interface debonding, and delamination are the major deterioration processes [35]. The resin degradation expands within resin micropores and microcracks, allowing water molecules and corrosive ions to enter the resin and fiber-resin interface quickly, eventually leading to the fiber-resin interface debonding [67,125]. Once the interface is damaged, water molecules and corrosive substances rapidly penetrate the interface, further exacerbating the damage of the interface [69,126]. Fig. 17(c) shows the SEM images of the interfacial debonding between fiber and resin.

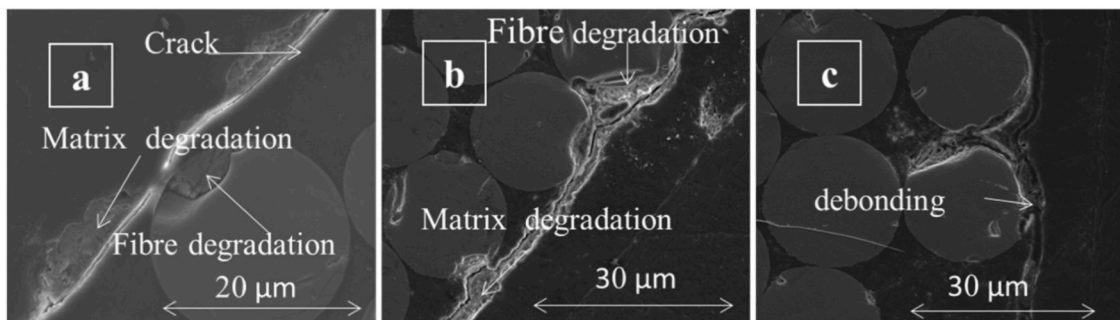


Fig. 17. SEM images of fiber-resin interface debonding: (a) matrix degradation, (b) fiber degradation, and (c) interface debonding [109].

3.4. Summary

When FRP bars are exposed to water or an alkaline environment, the resin initially degrades due to plasticizing and hydrolysis (see Section 3.1). The degradation of resin adversely affects the load transfer capability of FRP bars, lowering their tensile strength. After losing resin protection of the FRP bar, water and OH^- , accompanied by the macromolecular chains of resin, rapidly reach the fiber-resin interface. Fiber undergoes severe deterioration due to OH^- etching and Cl^- corrosion (basalt fiber). The fiber-resin interface loses its mechanical biting force and chemical bonding force. The failure of fiber and resin eventually leads to the deterioration of FRP bars. Thus, it can be deduced that the resin hydrolysis is the primary cause of the FRP bar damage, followed by the debonding of fiber-resin interface and fiber damage, and ultimately FRP bar failure [109], as illustrated in Fig. 18.

4. Service-life prediction models for the durability of FRP bars

To evaluate the long-term durability of FRP bars, it is necessary to carry out the long-life prediction based on the accelerated test results obtained in the short-term [127]. The prediction models based on the Arrhenius relation and refined models considering environment reduction factors are discussed in this section, and the prediction results are compared.

4.1. Basic models

At present, all commonly used service-life prediction models for FRP composites are based on the Arrhenius acceleration theory, which assumes that the degradation mechanism remains unchanged with increasing temperatures and aging time, but only the degradation rate increases. According to the Arrhenius acceleration theory, the relationship between the degradation rate of FRP materials and aging temperature in the three models can be expressed as follows [37].

$$K = A \exp\left(\frac{E_a}{R_T}\right) \quad (7)$$

where K is the degradation rate (MPa/d), A is the material constant and degradation process, E_a is the material activation energy (J/mol), R is the molar gas constant, 8.3143 (J/(mol·K)), and T is the ambient absolute temperature (K).

Equation (7) can be further expressed as

$$\ln\left(\frac{1}{K}\right) = \frac{E_a}{R} \cdot \frac{1}{T} - \ln A \quad (8)$$

In Equation (8), the degradation rate K is the reciprocal of the aging time required for a material property to degrade to a given retention rate. This Equation also shows a linear relationship between the logarithmic aging time ($1/K$) and reciprocal absolute temperature ($1/T$) with a slope of E_a/R . Based on Equation (7), the time-shift factor for the tensile strength to reach the same value at temperatures T_1 and T_0 can be estimated using Equation (9).

$$TSF = \frac{t_0}{t_1} = \frac{Y/k_0}{Y/k_1} = \frac{k_1}{k_0} = \frac{A \exp(-E_a/RT_1)}{A \exp(-E_a/RT_0)} = \exp\left[\frac{E_a}{R} \left(\frac{1}{T_0} - \frac{1}{T_1}\right)\right] \quad (9)$$

Researchers [30,73,128–130] have developed a variety of durability prediction models. Four types of the most widely used prediction models are considered and discussed in the following part.

The first model 1 proposed by Bank et al. [129] assumes a linear relationship between the logarithmic coordinate of aging time.

$$\text{Model1} : Y = \log(t) + b \quad (10)$$

where Y is the FRP mechanical properties retention rate (%), t is the aging time, and a and b are the constants of two regression equations. This model has successfully predicted the long-term tensile properties of FRP bars embedded in concrete [37,65,128]. However, the model is only a linear regression of the test results and does not give composite materials a precise degradation mechanism. In this model, the strength retention rate of the sample at no aging (i.e., at time 0) is infinite, which is obviously contrary to the actual condition.

Models 2 and 3 assume that the properties of FRP materials degrade rapidly at the initial aging, then gradually slow down and eventually plateau.

$$\text{Model2} : Y = 100 \exp\left(-\frac{t}{\tau}\right) \quad (11)$$

$$\text{Model3} : Y = (100 - Y_\infty) \exp\left(-\frac{t}{\tau}\right) + Y_\infty \quad (12)$$

where τ is the fitting parameter, and Y_∞ is the mechanical property retention rate (%) corresponding to FRP bars at the infinity aging. Phani and Bose [130] proposed these two models to predict the flexural strength degradation of CFRP composite laminates in a hot and humid

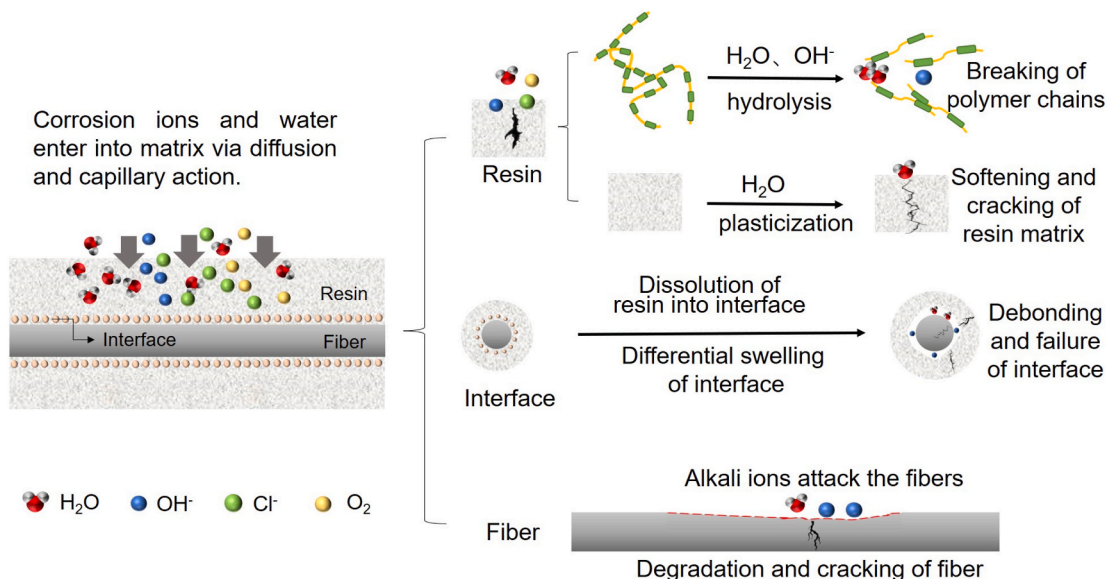


Fig. 18. Degradation mechanisms of FRP bars.

environment. After that, other researchers modified them to predict the long-term tensile properties of GFRP and BFRP bars in a concrete environment [73]. These two models provide a specific degradation mechanism; that is, the degradation of FRP material mainly derives from the debonding of the fiber-resin interface inside the material. The main difference between model 2 [73] and model 3 [130] is that the final FRP mechanical property retention tends to zero or Y_∞ . Model 2 and model 3 have been used in Refs. [11,36,73,96,131] and Ref [132], respectively. Wang et al. [37,111] compared these two models to obtain long-term prediction results and found that model 2 is much more conservative than model 3 for the FRP bar predictions.

The fourth "moisture absorption" model was suggested by Tannous et al. [133] as

$$Y = 100 \left(1 - \frac{\sqrt{2DCt}}{r_0} \right)^2 \quad (13)$$

where t is the exposure time, D is the diffusion coefficient, C is the solution concentration, and r_0 is the FRP bar radius. The diffusion model based on Fick's law is employed to build a relationship between strength retention and moisture absorption of FRP bars [128]. Model 4 has been applied in Ref. [71]. However, this model assumes that the affected area is entirely deteriorated, which may not represent the true condition of FRP bars. Moreover, determining D and C is rather complicated; thus, these drawbacks limit the use of this model [134].

4.2. Refined prediction method

Based on the deterioration mechanism of FRP bars subjected to harsh environments (as discussed in Sections 2 and 3) and existing models (Section 4.1), some researchers proposed a refined prediction model involving the effects of temperature fluctuation [135], concrete wrapping [134], design life (DL) [30], and environmental relative humidity (RH) [30]. Dong et al. [134] classified accelerated aging tests of FRP bars into three categories: type I, type II, and type III, as depicted in Fig. 19. Type I aging test with correction factors of α , β , and η represent bare FRP bars immersed in the liquid solution (usually simulated concrete pore solution). FRP bars wrapped in concrete immersed in the liquid solution are indicated by a type II test with correction factors of β and η . Type III testing with a correction factor of η implies FRP bars wrapped in concrete exposed to less than 100% RH, which is closer to the actual environment. Note that the temperature remains constant in

all three cases.

The three basic forms of water in concrete are capillary water, physically absorbed water, and interlayer water [136]. As a result, Huang et al. [30] proposed the correction factor β , assuming that it is equal to the ratio of the capillary and adsorbed water in concrete (Fig. 20). In addition, Huang et al. [137] investigated the correction factor β of FRP bars in China, while Dong et al. [134] suggested the correction factor α is 2.18 for bare FRP bars. The correction factor η based on the time shift factor (TSF) can be obtained by [135].

$$\eta = \frac{12}{\sum_{i=1}^{12} \exp \left[\frac{E_a}{R} \left(\frac{1}{T_{AT}} - \frac{1}{T_{MT}} \right) \right]} \quad (14)$$

where T_{AT} is the mean annual temperature ($^{\circ}\text{C}$) and T_{MT} is the monthly mean temperature ($^{\circ}\text{C}$).

According to the environmental reduction factors stated above, Huang and Aboutaha [30] proposed an improved prediction method based on model 1 as given by

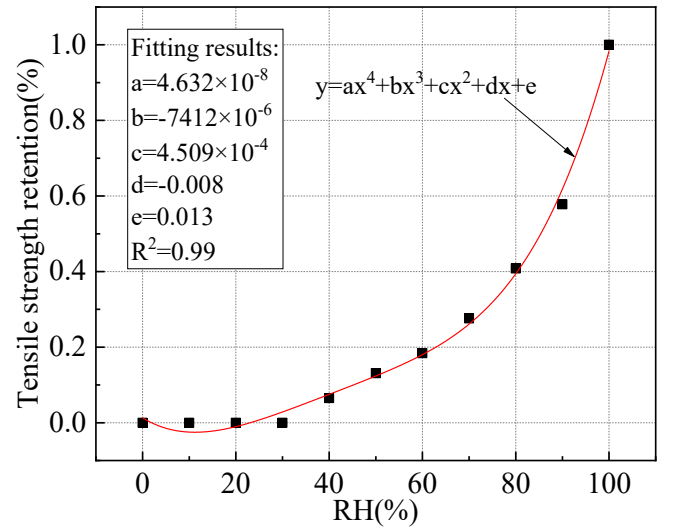


Fig. 20. Relationship between the correction factor and the RH (Adapted from Huang and Aboutaha [30]).

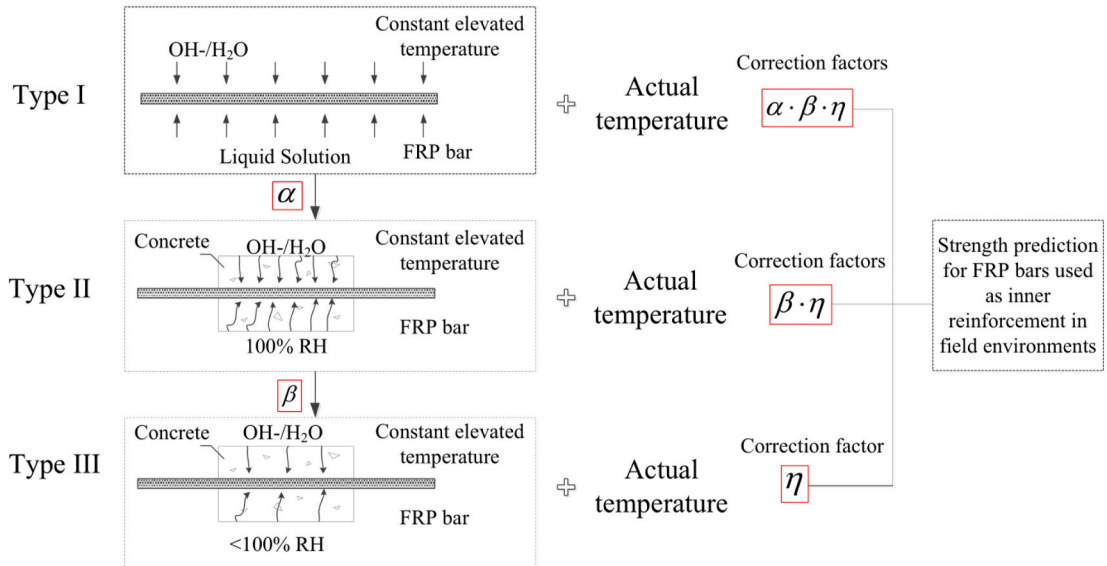


Fig. 19. Different types of aging tests and combinations of correction factors (α , β , and η are the correction factors of concrete cracking, environmental humidity, and actual temperature fluctuation, respectively) [134].

$$ERF = 1 - \beta \cdot \left[\Delta_1 + \rho \cdot \log \left(\frac{DL}{\alpha \cdot TSF \cdot \eta} \right) \right] \quad (15)$$

where ERF is the environment reduction factor, Δ_1 is the strength deterioration after one year of experimental exposure, ρ is the Arrhenius regression curve slope of model 1, and DL is the design life of FRP bars. Notably, the refined prediction method suggested by Huang and Aboutaha [30] is based on model 1, which is the linear regression curve of aging test results. However, linear regression curves of aging test measurements from different studies [37,63,138] are not parallel; a similar problem was reported in Ref. [73]. To overcome this drawback, refined models containing ERF based on models 2 and 3 are presented in Equations (16) and (17), respectively. There are yet no studies on refined models 2 and 3 available in the literature.

$$100 - \frac{(100 - Y)}{\alpha \cdot \beta} = 100 \exp \left(-\frac{t}{\tau} \right) \quad (16)$$

$$100 - \frac{(100 - Y)}{\alpha \cdot \beta} = (100 - Y_\infty) \exp \left(-\frac{t}{\tau} \right) + Y_\infty \quad (17)$$

4.3. A comparison of predictions based on existing models

Based on the test results of Wang et al. [37] and Wu et al. [73], the commonly used models 1, 2, 3, and refined model 2 are compared in this section (see Table 2). Representative cities of Moscow (55°50'N), Russia, Beijing (39°56'N), China, Wuhan (29°58'N), China, Miami (25°45'N), USA at different latitudes are chosen to predict the tensile strength retention of FRP bars. Moscow, Beijing, Wuhan, and Miami have mean annual temperatures of 5, 11.8, 22.3, and 24.4 °C, respectively.

Fig. 21 displays the prediction results of models 1, 2, and 3. According to ACI 440.1R-2015 [110], the suggested environmental reduction factor (C_E) for FRP bars in concrete exposed to earth and weather is 70%. The predicted time to reach 70% retention for BFRP bars exposed to SWSSC and alkaline solution in four cities is calculated using models 1, 2, and 3 (Fig. 21). Because higher latitudes have a lower temperature, longer exposure duration are expected to decline to 70% retention. When comparing Fig. 21(a) and (b) using experimental data of Wu et al. [73], the prediction results of model 2 are more conservative than model 1. Similarly, model 2 is more conservative than model 3 in Fig. 21(c) and (d), respectively. It should be noted that model 1 is not suitable for aging test results of Wang et al. [37] since regression curves are not parallel to each other, violating Arrhenius theory. Furthermore, the slope (E_a/R) of Arrhenius curves is 20753 using model 3 from the experimental data of Wu et al. [73], which is not achievable in practice as acceptable reported values of E_a/R are 1420, 7519, and 6696 in Refs. [36,73], and [111], respectively. Variation in E_a/R leads to varying degradation rates and mechanisms of FRP bars exposed to various environments [128]. It can be inferred that models 1 and 3 are not widely used for the aging test. Although model 2 is appropriate for the majority of experimental data, the prediction results are conservative; a similar result was observed in Refs. [37,73,111]. Thus, refined model 2 is required to obtain more accurate prediction results.

Based on the shortcomings of models 1, 2, and 3, a refined model 2

considering ERF outlined in Section 4.2 is used in this part. Wu et al. [139] evaluated the long-term durability of FRP bars embedded in concrete in Wuhan, China, and found 70.6% tensile strength retention after eight years of alkaline erosion. To compare the predicted results of refined model 2 and models 2 and 3 with the real data of Wu et al. [139], the city of Wuhan is chosen using the aging test results of Wang et al. [37], and the prediction results are shown in Fig. 22. Due to the accelerating test results of Wu et al. [139] obtained in the alkaline solution, correction factors of η and α are considered in refined model 2. After eight years, 75.9%, 40.3%, and 72.3% tensile strength retention are predicted using refined model 2, model 2, and model 3, respectively. Predictions of refined model 2 are quite close to the real results of Wu et al. [139], and model 2 is conservative. The slight difference in prediction results between refined model 2 and actual data can be attributed to harsh alkaline immersion corrosion and FRP bar disparity. Although the prediction results of model 3 are about 72.3% strength retention, closer to the real results of Wu et al. [139], the prediction results consistently exceeding 70.8% with aging time are not feasible in practical applications.

The models above may be used to determine the service life of FRP bars at any place. Although refined model 2 provides more accurate predictions than other models, most of them are based on the Arrhenius assumption, which is an idealized condition. Furthermore, the lack of FRP material data under service conditions makes it challenging to compare prediction results to FRP material in actual service life. Recently, a new prediction model based on the high nonlinear ability of artificial intelligence provided a new direction for prediction methods [140].

5. Recommendations for further research

- (1) Temperature, alkalinity, humidity, and high-stress level are the most critical factors affecting the FRP bar deterioration. Developing appropriate methods to mitigate moisture and alkali erosions is required to enhance the durability of FRP reinforced concrete structures successfully.
- (2) Combining two or three kinds of fiber in FRP bars may be performed to achieve high durability and high strength at a low cost using the benefits of corrosion resistance of carbon fiber and the relatively low cost of basalt and glass fibers. Moreover, resin modification should be done to improve the weatherability of the matrix to protect the fiber better. In addition, new material can be introduced to coat FRP bar surfaces to protect against moisture and alkali attacks.
- (3) To improve the waterproof performance of concrete to reduce water penetration, concrete surfaces can be coated with a deep penetration sealer, silane emulsion, or silane emulsion mixed inside the concrete. Decreasing the concrete alkalinity should be ensured to improve the durability of FRP bars effectively. The internal alkalinity of the concrete can be reduced by adjusting the binding material or rapid carbonization to avoid the alkaline corrosion of FRP bars.
- (4) Variations in alkalinity, internal humidity, and pore structures caused by the water-cement ratio, cover thickness, and supplementary cementitious materials in concrete can affect the durability of FRP bars. The deterioration of FRP bars wrapped in concrete needs to be studied further under the combined action of environmental factors and sustained loading with varying stress levels.
- (5) Most studies investigate accelerated corrosion in the laboratory environment, which differs from the field exposure environment. Therefore, future research should focus on the durability of FRP bars in the outdoor environment to obtain reliable data. Establishing the relationship between laboratory aging tests and field exposure environment is also required.

Table 2
Experimental tensile strength retention of BFRP bars.

References	Specimen type	Exposure solutions	Exposure temperature (°C)	Tensile strength retention (%)		
				21d	42d	63d
Wang [37]	BFRP bars	SWSSC (pH = 12.7)	32	99.3	93.8	97.9
			40	95.3	93.0	90.2
			55	89.7	81.1	77.4
Wu et al. [73]	BFRP bars	Alkaline (pH = 12.9–13.1)	25	99.1	98.2	94.0
			40	90.3	88.9	84.8
			55	82.0	76.8	68.5

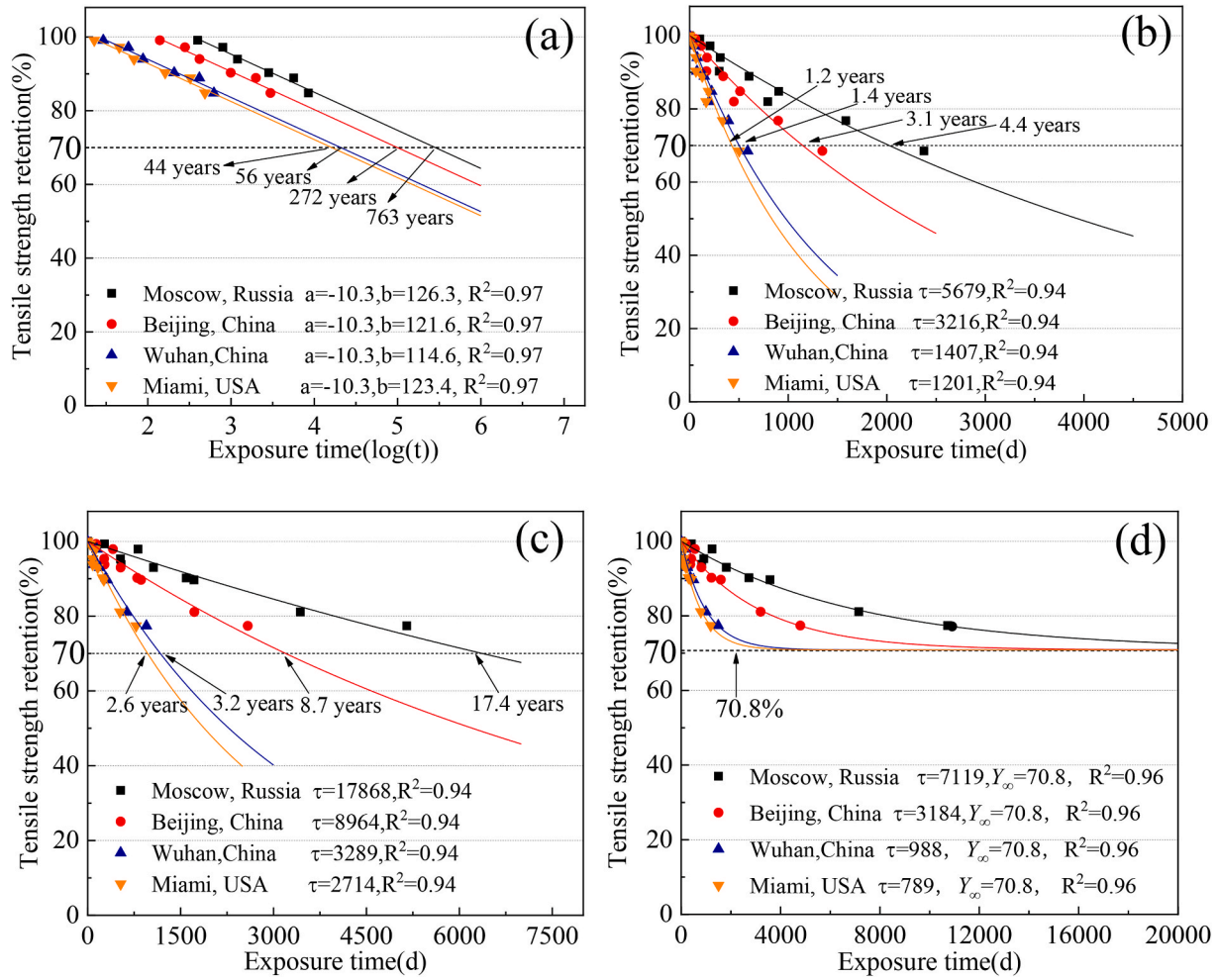


Fig. 21. Tensile strength retention of BFRP bars predicted using models 1, 2, and 3: (a)–(b) prediction results of models 1 and 2 using experimental data of Wu et al. [73], respectively, and (c)–(d) prediction results of models 2 and 3 using experimental data of Wang et al. [37], respectively.

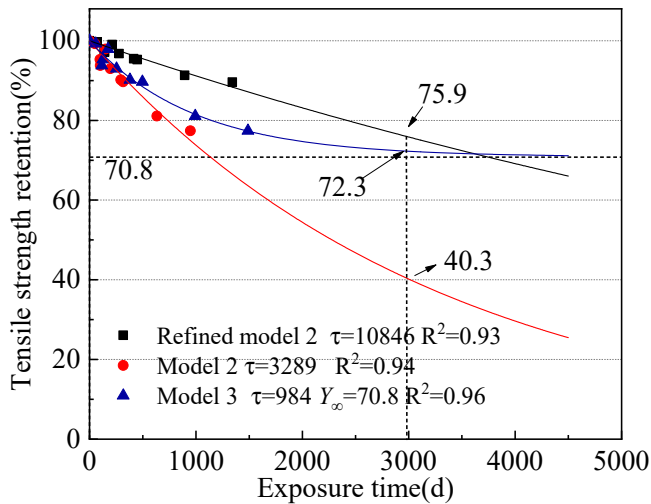


Fig. 22. A comparison of prediction results for BFRP bars in Wuhan, China using refined model 2, model 2, and model 3.

influence of key environmental factors and their associated degradation mechanisms in FRP bars should be integrated into one long-term prediction model for obtaining comprehensive information.

6. Conclusions

- (1) The long-term performance of FRP bars can be affected by both the exposure condition and their characteristics. The deterioration rate of FRP bars generally increases with the alkalinity of the solution. The durability performance of the CFRP is better than that of the GFRP and BFRP. The FRP bars prepared with vinyl ester have better long-term performance than those prepared with epoxy and polyester resins.
- (2) The alkaline erosion of FRP bars is the main cause of their deterioration. The durability performance of FRP bars in seawater is almost identical to that in tap water. The influence of wet-dry and freeze-thaw cycles on the long-term performance of FRP bars is insignificant due to the less time for moisture erosion.
- (3) The deterioration of FRP bars immersed in the simulated alkaline solution does not accurately represent the real concrete environment. FRP bars in moist concrete have higher strength than FRP bars soaked in an alkaline solution. In addition, the protective effect of concrete for FRP bars differs from that of steel bars; both the thicker concrete cover and lower water-cement ratio

accelerate FRP degradation due to increased interior alkalinity of concrete.

- (4) The mechanical performance of FRP bars is highly temperature-sensitive. An increase in the mechanical performance of the FRP bar can be observed under the temperature range of -100°C to 0°C . The tensile strength drops rapidly when the temperature reaches 300°C and loses most of the loading capacity after exceeding 500°C .
- (5) The combined action of the environment and sustained loading can significantly affect the failure mode of FRP bars. The diffusion of corrosive ions dominates the deterioration under the low-stress level (25%–30% of ultimate strength). The degradation of the FRP bar is mainly controlled by crack propagation under medium stress levels (30%–50% of ultimate strength). The failure mode becomes brittle fractures when the applied load exceeds 50% of ultimate strength.
- (6) Current long-term prediction models for FRP bars are based on the Arrhenius acceleration theory and are conservative compared to field data. The refined prediction model considering the environmental correction factors (relative humidity, dynamic temperature, and protection of concrete) is more suitable for predicting the long-term performance of FRP bars.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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